

Rhythm Gate: Invisible Conversations in the Elevator — Echoes of Material, Behavior, Memory and Transformation

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Abstract

RhythmGate reimagines the elevator as a spatial medium for sensing and expressing the unspoken codes of shared space. By combining pressure-sensitive flooring with overhead visual sensing, the system captures bodily presence and spatial movement in real time. These inputs are processed to classify position and trigger one of five pre-recorded human voices, selectively layered into a responsive soundscape. As riders shift their stance, they “cut” between temporal zones mapped across the floor, composing a living acoustic montage. The resulting auditory feedback surfaces subtle social dynamics—such as avoidance, boundary-setting, or latent tension—and gently dissolves them. In doing so, RhythmGate transforms a routine ascent into a fleeting, multisensory conversation among strangers, system, and space. See artwork video.

CCS Concepts

• **Applied computing** → **Media arts; Sound and music computing; • Human-centered computing** → *Interaction design theory, concepts and paradigms.*

Keywords

Interactive Art, Sound Installation, Multisensory Interface, Spatial Montage, Proxemics

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1 Introduction

Elevators epitomise the public “transition zone”: strangers are compressed into a moving box for seconds, performing a mute choreography of gaze avoidance and territorial micro-shifts [1, 3]. These tacit routines keep the ride friction-free, yet they seal social tension inside the cabin the instant the doors close. While interactive art has repeatedly sonified hidden urban data—heartbeat, pollution, crowd density—to expose latent patterns [5, 6], the elevator’s peculiar mix of brevity, confinement, and hierarchy remains largely unexplored.

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RhythmGate positions the elevator as both screen and score, spatially unfolding time the way a film strip lays out its frames. Five floor zones stand for five temporal slices; stepping from one to another “cuts” to a new moment, and the cabin’s interior becomes a living montage of otherwise invisible instants. A multisensory pipeline—pressure-sensor carpet, overhead computer vision, and zone-specific vocal cues—audibly surfaces these micro-negotiations, inviting riders to move, listen, and, by extension, converse. When voices accumulate, the work breaks the standard spell of elevator silence; when real speech emerges, the system’s prompts recede, yielding the stage to human exchange. The accompanying video mirrors this arc: from a single quiet frame to twenty-five concurrent windows and back to calm, it compresses the social crescendo within a five-minute loop. This paper contributes:

- (1) **Time-space unfolding framework.** Reframe the elevator floor as temporal slices, extending *spatial montage* into an immersive sonic setting.
- (2) **Real-time dual-modal sensing.** Lightweight pressure sensing plus overhead vision robustly track multi-user zone shifts in a dynamic cabin.
- (3) **Density-adaptive zone-sound mapping.** A loudness normalised layering scheme scales from single rider to full capacity and fades when natural speech appears.
- (4) **Behavioural effect.** Field notes show the soundscape nudges riders from silent avoidance toward casual conversation.

2 Conceptual Framework & Artistic Intervention

RhythmGate treats space as a timeline that can be unrolled, frame by frame, across the floor of a lift. Drawing on Manovich’s concept [7] of *spatial montage* and Elcott’s readings of expanded cinema [2], the piece slices five temporal moments onto five fixed zones; every step is a physical “cut” to a new slice of time.

The elevator’s notorious etiquette of silence and gaze-aversion is therefore confronted not by imagery but by sound—a medium that permeates closed volumes without demanding visual attention. Each zone is linked to a distinct vocal timbre, so that the rider’s weight literally “presses” a voice into being. The dual-modal sensing pipeline is not an engineering flourish but a material gesture: force sensors capture the mass of presence, while computer vision captures its outline; the two together carve an acoustic relief that riders inhabit and reshape in real time.

In the accompanying video, a single quiet frame gradually multiplies into twenty-five concurrent windows before dissolving back into stillness, mirroring the sonic arc from solo line to polyphonic density and, finally, to collective release. Technology thus functions

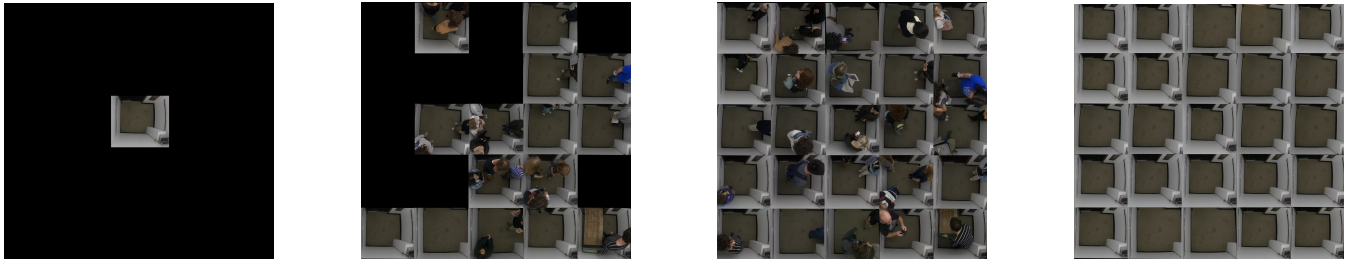


Figure 1: *RhythmGate* video arc. The projection begins with a single quiet frame and gradually expands into a growing grid of elevator scenes. As the number of frames increases, so does the sonic density—layering pre-recorded vocal cues with real human sounds as passengers shift from silence to spontaneous exchange. Once all 25 windows are revealed, the visual composition reaches its most intricate form, yet the soundscape fades back to calm: the elevator empties, and the space exhales.

as both chisel and stage, enabling an auditory intervention that momentarily fractures the choreography of elevator silence.

3 Interaction Scenario

The cabin floor is partitioned into five fixed zones—front-left, front-right, centre, rear-left, rear-right—each bound to a unique pre-recorded human voice (Voice 1–5). A large force-sensing carpet (2 kg resolution) streams weight vectors to an Arduino Touch Board, while a ceiling-mounted GoPro delivers 4K 60 fps video to a concealed MacBook that localises bodies via a lightweight YOLO-v8 model. The two streams are fused at 140 ms end-to-end latency, forming a live, zone-indexed “spatial score” in which movement, like editing, unfolds time across space.

Solo entry — the first frame. When a lone passenger crosses the threshold, the system fades in the voice attached to the occupied zone, writing the journey’s opening “frame” as a single audible line. If the rider drifts to a new zone, a 250 ms cross-fade transfers the melody, turning every footstep into a cut from one temporal slice to the next.

Co-presence layering — the montage builds. Subsequent entrants activate the voices of their landing zones; all tracks layer without a hard limit, each normalised in real time to preserve speech intelligibility. The elevator thus accumulates concurrent voices as a video would accumulate frames, weaving an ever denser polyphony that mirrors the growing social density. Because the vocal timbres are intentionally light and open-voiced, they slip beneath spontaneous conversation; passengers who begin to chat soon hear their own speech bury the system’s prompts, signalling that human dialogue has replaced enforced silence.

Departure — the fade-out. As riders exit, vacated zones release their voices in reverse order of occupation, thinning the sound field until only one line, and finally none, persists. The cabin returns to quiet with no closing cue, leaving the residual ambience to “breathe” before the doors open again.

Through this sequence the installation stages an audible arc—calm → polyphonic density → calm—while revealing how fleeting co-presence can be spatially and sonically composed. Each ride becomes a live montage of otherwise invisible moments, and sound, rather than sight, acts as the medium that dissolves elevator awkwardness into shared, if temporary, conversation.

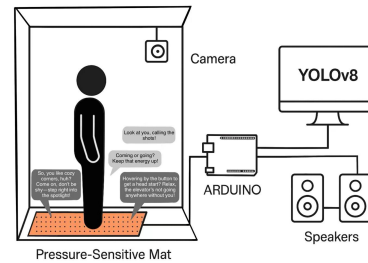


Figure 2: System pipeline of *RhythmGate*. A pressure-sensitive mat detects footfall and zone shifts, while an overhead camera feeds a YOLOv8 model for top-view localisation. Both streams are fused via an Arduino controller; the resulting zone events cue the corresponding pre-recorded voices, mixed and played through stereo speakers inside the cabin.

4 System Overview

Figure 2 illustrates a lean, two-sensor loop. A large FSR carpet plugs into an Arduino Touch Board, which streams weight data to a concealed MacBook over USB. An overhead GoPro Hero 11 supplies a 4 K feed that a lightweight YOLO-v8 [4] model fine-tuned on VisDrone [8] converts into top-view person coordinates. Either source—pressure spike or body centroid—flags one of five floor zones; the MacBook passes the active zone ID to a Max/MSP patch that cues the matching 48 kHz voice clip. All clips mix in real time with RMS auto-levelling, then play through a pair of 20 W powered bookshelf speakers. Sensor-to-sound latency averages 120 ms, well inside the 200 ms window for synchronous feedback. The entire rig draws ≤ 500 W and fits inside a 1.5 m $\times$ 1.5 m mock elevator frame.

5 Conclusion

RhythmGate casts the elevator as a cinematic strip unspooled underfoot, its five vocal zones translating passing bodies into audible edits in real time. By coupling minimal force sensing with overhead vision, the work treats technology as sculptural matter: it chisels silence into speech, hesitation into melody, and shared confinement into a brief social score. In doing so, it demonstrates how spatialised sound can not only reveal the micro-politics of proxemics but momentarily dissolve them, turning a routine ascent into a collective, time-layered conversation. We hope this approach—time unfolded through multisensory feedback—encourages future artists to re-author the infrastructures that silently choreograph public life.

References

- [1] Michael Dellwing. 2022. Relations in Public. Microstudies of the Public Order. In *Goffman-Handbuch: Leben–Werk–Wirkung*. Springer, 323–329.
- [2] Noam M Elcott. 2016. Artificial darkness: An obscure history of modern art and media. In *Artificial Darkness*. University of Chicago Press.
- [3] Edmund T Hall and Edward T Hall. 1966. *The hidden dimension*. Vol. 609. Anchor.
- [4] Muhammad Hussain. 2024. Yolov5, yolov8 and yolov10: The go-to detectors for real-time vision. *arXiv preprint arXiv:2407.02988* (2024).
- [5] Golan Levin, Gregory Shakar, Scott Gibbons, Yasmin Sohrawardy, Joris Gruber, Erich Sendlak, Gunther Schmidl, Joerg Lehner, and Jonathan Feinberg. 2001. Dialtones-a telesymphony.
- [6] Rafael Lozano-Hemmer. 2013. Voice Tunnel.
- [7] Lev Manovich. 2002. The language of new media.
- [8] Pengfei Zhu, Longyin Wen, Dawei Du, Xiao Bian, Heng Fan, Qinghua Hu, and Haibin Ling. 2021. Detection and tracking meet drones challenge. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 11 (2021), 7380–7399.